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Assessment of Retinopathy of Prematurity Regression and Reactivation Using an Artificial Intelligence–Based Vascular Severity Score

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Abstract

IMPORTANCE One of the biggest challenges when using anti–vascular endothelial growth factor (VEGF) agents to treat retinopathy of prematurity (ROP) is the need to perform long-term follow-up examinations to identify eyes at risk of ROP reactivation requiring retreatment.

OBJECTIVE To evaluate whether an artificial intelligence (AI)–based vascular severity score (VSS) can be used to analyze ROP regression and reactivation after anti-VEGF treatment and potentially identify eyes at risk of ROP reactivation requiring retreatment.

DESIGN, SETTING, AND PARTICIPANTS This prognostic study was a secondary analysis of posterior pole fundus images collected during the multicenter, double-blind, investigator-initiated Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity (CARE-ROP) randomized clinical trial, which compared 2 different doses of ranibizumab (0.12 mg vs 0.20 mg) for the treatment of ROP. The CARE-ROP trial screened and enrolled infants between September 5, 2014, and July 14, 2016. A total of 1046 wide-angle fundus images obtained from 19 infants at predefined study time points were analyzed. The analyses of VSS were performed between January 20, 2021, and November 18, 2022.

INTERVENTIONS An AI-based algorithm assigned a VSS between 1 (normal) and 9 (most severe) to fundus images.

MAIN OUTCOMES AND MEASURES Analysis of VSS in infants with ROP over time and VSS comparisons between the 2 treatment groups (0.12 mg vs 0.20 mg of ranibizumab) and between infants who did and did not receive retreatment for ROP reactivation.

RESULTS Among 19 infants with ROP in the CARE-ROP randomized clinical trial, the median (range) postmenstrual age at first treatment was 36.4 (34.7-39.7) weeks; 10 infants (52.6%) were male, and 18 (94.7%) were White. The mean (SD) VSS was 6.7 (1.9) at baseline and significantly decreased to 2.7 (1.9) at week 1 ($P < .001$) and 2.9 (1.3) at week 4 ($P < .001$). The mean (SD) VSS of infants with ROP reactivation requiring retreatment was 6.5 (1.9) at the time of retreatment, which was significantly higher than the VSS at week 4 ($P < .001$). No significant difference was found in VSS between the 2 treatment groups, but the change in VSS between baseline and week 1 was higher for infants who later required retreatment (mean [SD], 7.8 [1.3] at baseline vs 1.7 [0.7] at week 1) vs infants who did not (mean [SD], 6.4 [1.9] at baseline vs 3.0 [2.0] at week 1). In eyes requiring retreatment, higher baseline VSS was correlated with earlier time of retreatment (Pearson $r = -0.9997$; $P < .001$).

(continued)

Key Points

Question Can an artificial intelligence–based vascular severity score (VSS) be used for analyzing regression and reactivation of retinopathy of prematurity after anti–vascular endothelial growth factor (VEGF) treatment?

Findings In this prognostic study of 19 infants with retinopathy of prematurity who participated in the Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity randomized clinical trial, VSS decreased rapidly after anti-VEGF treatment. In eyes requiring retreatment, VSS increased again at the time of retreatment; these eyes also displayed a greater change in VSS between baseline and week 1, before the need for retreatment became clinically apparent.

Meaning These findings suggest that VSS after anti-VEGF treatment correlates with clinical disease severity and that the initial response in VSS to anti-VEGF therapy (from baseline to week 1) may help identify eyes that will later require retreatment.

+ Supplemental content

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Abstract (continued)

CONCLUSIONS AND RELEVANCE In this study, VSS decreased after ranibizumab treatment, consistent with clinical disease regression. In cases of ROP reactivation requiring retreatment, VSS increased again to values comparable with baseline values. In addition, a greater change in VSS during the first week after initial treatment was found to be associated with a higher risk of later ROP reactivation, and high baseline VSS was correlated with earlier retreatment. These findings may have implications for monitoring ROP regression and reactivation after anti-VEGF treatment.

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Introduction

Since retinopathy of prematurity (ROP) was first described by Terry[1] in 1942, both understanding of the disease and treatment options have continuously developed. With laser coagulation and anti–vascular endothelial growth factor (VEGF) drugs, 2 treatment approaches have evolved as the current standard of care for ROP.[2-5] Advantages of anti-VEGF drugs over laser coagulation include the potential for complete vascularization of the peripheral retina and a lower risk of myopia.[2,3,6-8] On the other hand, late recurrences can occur after anti-VEGF treatment that need to be detected using a close and long-term follow-up regimen.[9-11] Thus, with the advent of anti-VEGF drugs for ROP care, a second screening phase begins after treatment: screening for ROP reactivation requiring retreatment.[12]

To improve the quality and accessibility of primary ROP screening (ie, before treatment is given), several tools using artificial intelligence (AI) have been developed. Most of these tools focus on posterior pole vascular changes that are associated with ROP reactivation requiring retreatment. The International Classification of Retinopathy of Prematurity refers to such posterior pole vascular changes as the spectrum of plus disease (with plus disease defined as abnormal tortuosity and dilation of the posterior retinal blood vessels).[12] One of the established AI tools to quantify posterior pole vascular changes in ROP is the Images and Informatics in Retinopathy of Prematurity (i-ROP) deep learning (DL) algorithm.[13] Several ROP screening studies have found that the i-ROP algorithm has the ability to detect plus disease with a reliability that is comparable with human ROP experts[14,15] and therefore has potential to detect ROP reactivation requiring retreatment.[16-18] The output of the i-ROP DL system can be converted to a vascular severity score (VSS) reflecting the spectrum of plus disease and has been found to be useful for primary ROP screening, monitoring of disease progression, and response to treatment.[19-21]

In the current study, we examined the still-unanswered question regarding whether the VSS might be useful not only for primary ROP screening but for secondary ROP screening (ie, screening for ROP reactivation after anti-VEGF treatment). This analysis was conducted using the complete image set from the Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity (CARE-ROP) study, a multicenter randomized clinical trial comparing 2 different doses of ranibizumab for the treatment of ROP.[2,11]

Methods

This prognostic study was a secondary analysis of posterior pole fundus images collected during the prospective multicenter double-blind phase 2 CARE-ROP randomized clinical trial, which screened and enrolled infants between September 5, 2014, and July 14, 2016. The current analyses of VSS were performed between January 20, 2021, and November 18, 2022. This study followed the Standards for Reporting of Diagnostic Accuracy (STARD) reporting guideline for prognostic studies.[22] The CARE-ROP study was approved by the institutional review boards at the coordinating center (University Medical Center Freiburg) and each of the 9 study centers and was conducted in

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accordance with the Declaration of Helsinki.[23] Written informed consent was obtained from parents of all enrolled infants.

Data Set

All retinal images and clinical information for the current study were extracted from the CARE-ROP trial. Between September 2014 and July 2016, 20 infants were screened according to current German guidelines: (1) gestational age (GA) of less than 32 weeks or birth weight of less than 1500 g or (2) GA of 32 weeks or greater but with more than 3 days of oxygen supplementation or relevant comorbidities. The first ROP screening was at 6 weeks' postnatal age but not before 31 weeks' 0 days' GA. A total of 19 infants with bilateral ROP in zone 1 (stages 1 or 2 with plus disease, stage 3 with or without plus disease, or aggressive posterior ROP) or posterior zone 2 (stage 3 with plus disease or aggressive posterior ROP) were enrolled in the trial and received either 0.12 mg or 0.20 mg of ranibizumab.[2]

The CARE-ROP was also the first trial that allowed reinjection of the same study dose; if ROP status initially improved but ROP reactivated after at least 28 days posttreatment to a stage requiring retreatment, a reinjection of the same ranibizumab dose was permitted without unblinding the physician. If the disease did not initially improve after treatment or reactivated within 28 days after treatment, the treating physician was unblinded and patients received rescue treatment. The primary end point of the CARE-ROP study was the number of infants who did not require rescue therapy until 24 weeks after baseline. In the CARE-ROP trial, 8 eyes (4 eyes in each treatment arm) received per protocol retreatment, and 2 infants (1 eye each) needed rescue treatment. Further information about study design, results of the core study, and 2-year follow-up have been published previously.[2,11]

At predefined time points during the study (baseline [before injection] and 1 week, 4 weeks, and 24 weeks after treatment), ROP was digitally documented using an ophthalmic camera (RetCam; Natus Medical). Additional images were obtained from retreated infants at the time of retreatment as well as 1 week and 4 weeks after retreatment. Per protocol, centers were required to obtain 10 images per eye at each visit (details are provided in eFigure 1 in Supplement 1). A full image set was not always possible to obtain, but a total of 1413 images were acquired during the trial. After manual exclusion of 367 images due to poor quality or missing optic nerve head, the final data set for the current study contained 1046 images (details are provided in eFigure 2 in Supplement 1).

Image Grading

All images were graded using the i-ROP[13] computer-based image analysis system on a scale of 1 (normal retinal vasculature) to 9 (worst vascular abnormality).[19] This DL system had been trained on a reference standard diagnosis[17] to identify vascular characteristics of plus disease around the posterior pole as normal, preplus, or plus disease.[13,15]

Statistical Analysis

All images were analyzed using the i-ROP DL algorithm, with the output of a VSS (range, 1-9) for each image. The median VSS (evaluated using methods previously published[16]) of all images available for an eye per examination was determined separately for the left and right eye. Two eyes of 2 patients who required rescue treatment were excluded from all analyses (eFigure 2 in Supplement 1).

Using generalized estimating equations (GEEs) to account for intereye correlations and Bonferroni correction, we first evaluated whether there were significant differences between VSS in the 2 treatment groups (0.12 mg vs 0.20 mg of ranibizumab) at each time point. Subsequently, differences between the mean VSS of all eyes at different time points were compared using GEEs and multiple comparisons (Tukey contrasts) at baseline, week 1, week 4, week 24, and immediately before retreatment (if applicable). With Bonferroni correction, 2-sided *t* tests were used to compare VSS differences between ranibizumab dose groups and between infants who received retreatment vs those who did not at each time point. Pearson correlation coefficients were calculated to assess

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the correlations between VSS at baseline and (1) VSS at the time of retreatment and (2) time until retreatment. A GEE was also used to analyze the change in VSS from baseline to week 1 between infants who received retreatment and those who did not.

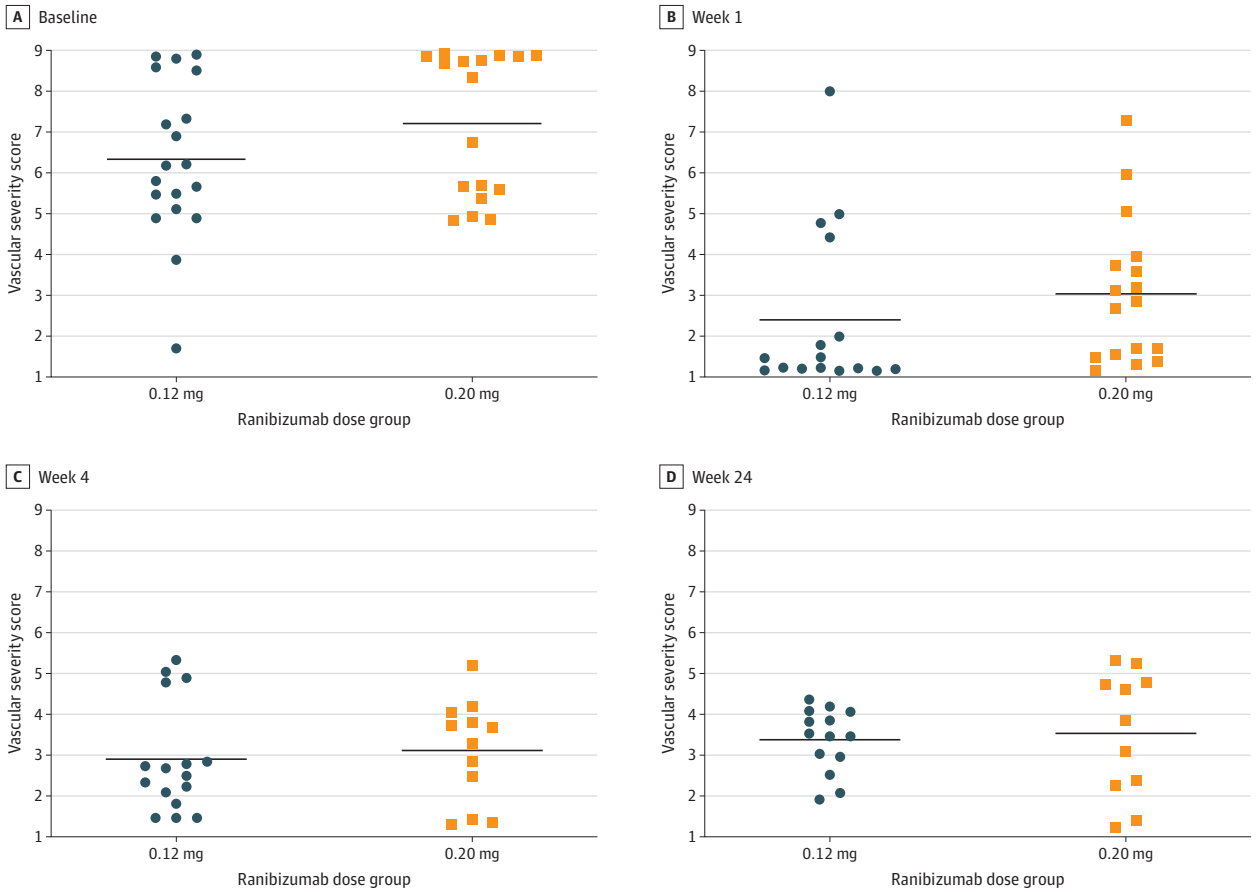
Analyses were performed using IBM SPSS Statistics software, version 28 (IBM Corporation), and graphic images were produced using Prism software, version 8.4.3 (GraphPad Software). The threshold for statistical significance was 2-sided $P \leq .05$.

Results

Comparison of 2 Different Doses of Ranibizumab

Among 19 infants with ROP in the CARE-ROP randomized clinical trial, the median (range) postmenstrual age at first treatment was 36.4 (34.7-39.7) weeks; 10 infants (52.6%) were male, and 18 (94.7%) were White. A total of 10 infants received treatment with 0.12 mg of ranibizumab, and 9 infants received treatment with 0.20 mg of ranibizumab. In the 0.12 mg group vs the 0.20 mg group, the median (range) GA at birth was 24.6 (22.9-29.6) weeks vs 24.7 (23.1-27.3) weeks, the median (range) postmenstrual age at first treatment was 36.8 (34.7-39.7) weeks vs 36.0 (35.1-39.6) weeks, and the median (range) birth weight was 555 (395-935) g vs 595 (360-1075) g.[2] The distribution of VSS in the 2 treatment groups (0.12 mg vs 0.20 mg of ranibizumab) at each time point is shown in **Figure 1**. For the 0.12 mg group vs the 0.20 mg group, the mean (SD) VSS was 6.3 (1.9) vs 7.2 (1.8) at

Figure 1. Comparison of Vascular Severity Scores Between Ranibizumab Dose Groups



Eyes that did and did not receive retreatment were included in this analysis; 2 eyes of 2 patients who required rescue treatment were excluded. Each marker represents the vascular severity score of 1 eye (using the median of all images available for an eye per visit). Black horizontal lines represent the mean values across the respective groups.

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baseline (Figure 1A), 2.4 (2.0) vs 3.0 (1.8) at week 1 (Figure 1B), 2.9 (1.3) vs 3.0 (1.3) at week 4 (Figure 1C), and 3.4 (0.8) vs 3.6 (1.4) at week 24 (Figure 1D). No significant difference in VSS scores between the 2 doses over time was found (eg, 0.12 mg vs 0.20 mg group from baseline to week 24: $P = .27$). The 2 treatment groups were combined for all subsequent analyses.

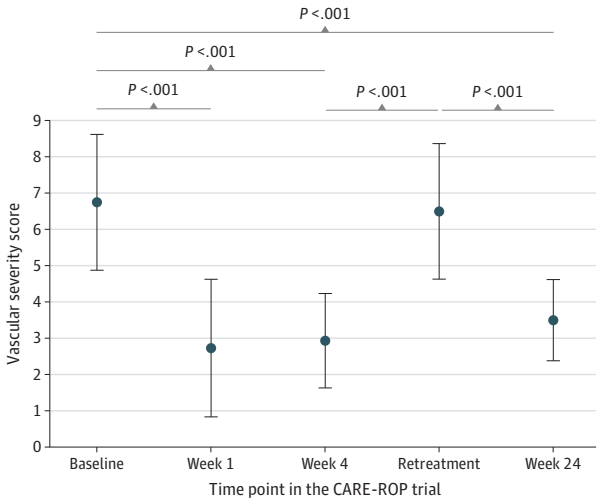
ROP Severity Before and After Treatment

The mean (SD) VSS for each participant in the CARE-ROP study before and after treatment and at the time of retreatment (if needed) is shown in **Figure 2**. At baseline, the mean (SD) VSS among 36 eyes was 6.7 (1.9), which significantly decreased to 2.7 (1.9) among 33 eyes at week 1 ($P < .001$) and 2.9 (1.3) among 29 eyes at week 4 ($P < .001$). The VSSs at week 1 vs week 4 were not significantly different from one another ($P = .98$). Eight eyes required retreatment at a mean (SD) of 10 (3.5) weeks (range, 7.1-15.0 weeks). At the time of retreatment, the mean (SD) VSS for these eyes was 6.5 (1.9), which was significantly higher than the VSS at the 4-week time point ($P < .001$). At the end of the trial (week 24), the mean (SD) VSS among 27 eyes was 3.5 (1.1), which was significantly lower than the VSS at baseline ($P < .001$) and at the time of retreatment ($P < .001$) but was not significantly different from the VSS at week 1 ($P = .39$) or week 4 ($P = .57$).

VSS for Retreated Eyes

The trajectory of the eye-level VSS (8 eyes) for the 4 infants who required retreatment over the course of the CARE-ROP trial is shown in **Figure 3**A. Qualitatively, the clinician's decision to provide retreatment appeared to correspond to increases in the VSS in all cases, although the magnitude of the increase varied, as did the period until reactivation. Between week 4 and retreatment, the mean (SD) VSS significantly increased from 3.0 (1.0) to 6.5 (1.9; $P = .001$). One infant (represented by the dark blue lines in Figure 3A) required 2 reinjections, both of which coincided with an increase in VSS. There was also a significant correlation (Pearson $r = 0.9110$; $P = .002$) between the VSS at the time of baseline treatment and the VSS at the time of retreatment for each eye (ie, infants who had higher VSS at the time of baseline treatment generally had an equally large increase in VSS before retreatment and vice versa). In addition, we observed a significant correlation (Pearson $r = -0.9997$; $P < .001$) between VSS at baseline and the time until retreatment became necessary; the higher the VSS at baseline, the sooner ROP reactivation requiring retreatment occurred (Figure 3B).

Figure 2. Development of the Vascular Severity Score During the Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity (CARE-ROP) Trial



Baseline represents the severity of retinopathy of prematurity before the first injection of ranibizumab. Retreatment represents the time point immediately before a reinjection of ranibizumab was administered for reactivation of retinopathy of prematurity that required retreatment. A total of 36 eyes were included at baseline, 33 eyes at week 1, 29 eyes at week 4, 8 eyes at retreatment, and 27 eyes at week 24. Variations in the number of eyes included over the study period were due to missing data at specific time points. Dots and whiskers indicated mean and SD.

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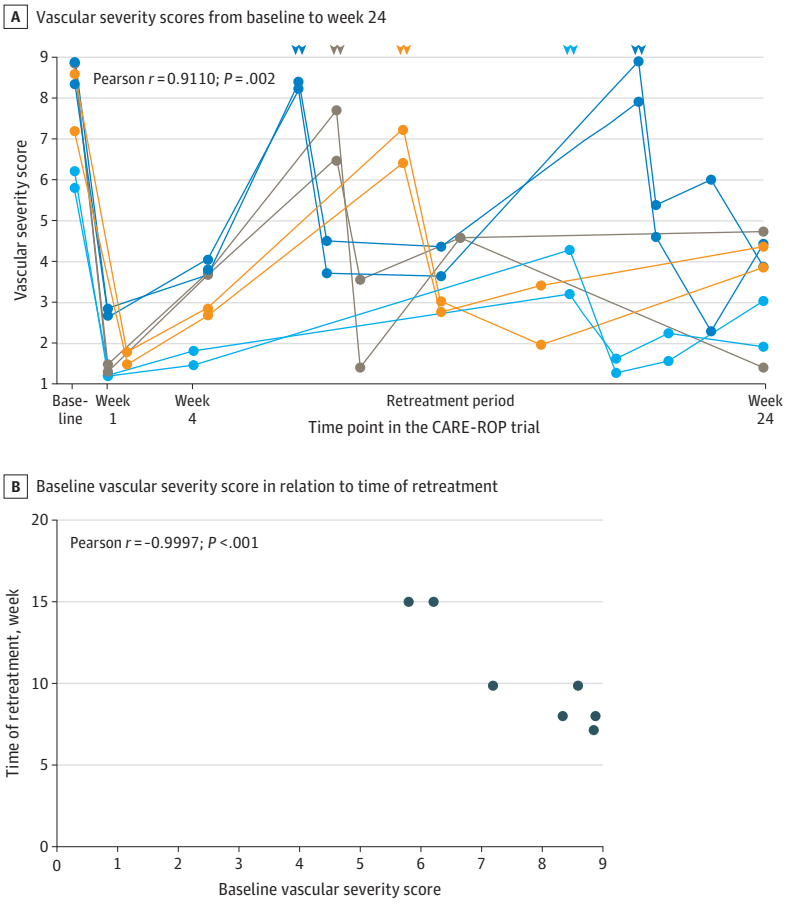
Changes in VSS in the Retreatment vs No Retreatment Group

In the CARE-ROP trial, retreatment was defined as a recurring need for treatment at least 4 weeks after injection and after an initial response to treatment. In the current analysis, we investigated whether VSS may be used early in the disease to estimate the need for later retreatment (**Figure 4**). For the group who did not receive retreatment vs the group who did, the mean (SD) VSS was 6.4 (1.9) vs 7.8 (1.3) at baseline (P = .05) (Figure 4A), 3.0 (2.0) vs 1.7 (0.7) at week 1 (P = .03) (Figure 4B), 2.9 (1.4) vs 3.0 (1.0) at week 4 (P = .87) (Figure 4C), and 3.5 (1.1) vs 3.5 (1.2) at week 24 (P = .87) (Figure 4D). The change in VSS from baseline to week 1 (**Figure 5**) was significantly larger in infants who later required retreatment compared with those who did not (P < .001).

Discussion

This prognostic study was a post hoc analysis of the CARE-ROP randomized clinical trial. It was a pilot study to evaluate the potential role of an AI-derived screening method in identifying disease reactivation after anti-VEGF treatment for ROP. The key findings of this analysis were (1) VSS corresponded well with clinical disease evolution in response to treatment and the need for retreatment; (2) the magnitude of VSS change during the first week after anti-VEGF treatment was associated with a higher probability of later requiring retreatment; (3) in eyes requiring retreatment, high VSS at baseline was associated with earlier retreatment; and (4) the 2 ranibizumab doses did not differ regarding VSS change after treatment.

Figure 3. Vascular Severity Scores in Eyes Requiring Retreatment

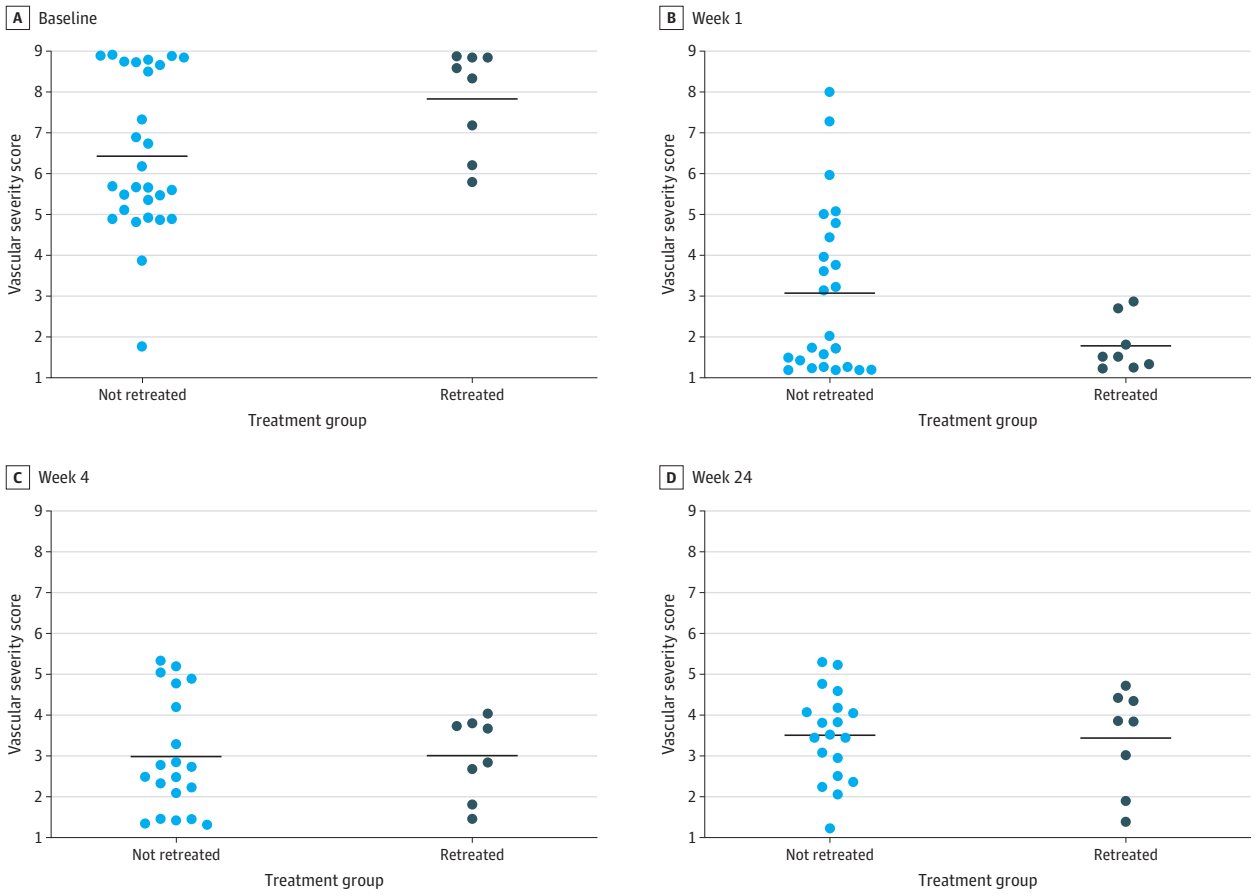


A, Each colored line represents the vascular severity score of the left or right eye of 1 infant requiring retreatment (8 eyes of 4 infants, with different colors representing different individual infants). The arrowheads represent the time points of reinjection with ranibizumab. One infant (dark blue lines) required 2 reinjections in both eyes. The x-axis is not to scale.

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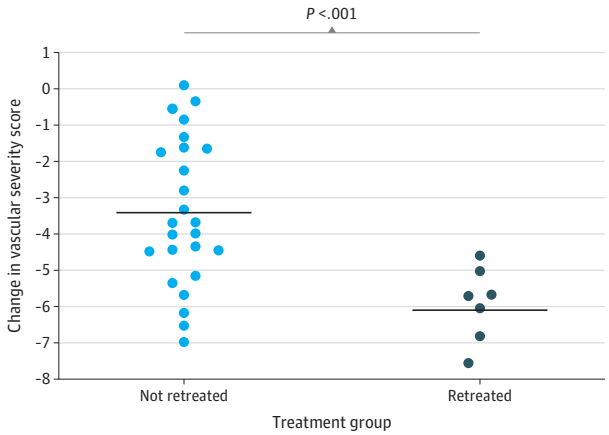
Previous work has described the use of i-ROP VSS for the assessment of overall ROP severity,[15,19] screening for type 1 ROP,[16-18] monitoring of disease progression,[20,24] and monitoring of initial response to treatment after laser or anti-VEGF therapy.[21] The results of our current analysis suggest that VSS may be useful for screening of ROP reactivation requiring retreatment after anti-VEGF therapy. Our results

Figure 4. Comparison of Vascular Severity Scores Between Eyes With vs Without Retreatment



Two eyes of 2 patients who required rescue treatment were excluded. Each dot represents the vascular severity score of 1 eye (using the median of all images available for an eye per visit). Black horizontal lines represent the mean values across the respective groups.

Figure 5. Change in Vascular Severity Scores Between Baseline and Week 1



Each dot represents the vascular severity score of 1 eye (using the median of all images available for an eye per visit). Black horizontal lines represent the mean values across the respective groups.

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revealed that baseline VSS values (before anti-VEGF treatment) were significantly higher compared with VSS values at 1 week after treatment and that, in each case of ROP reactivation requiring retreatment, VSS increased around the time when the CARE-ROP investigators clinically noted ROP reactivation requiring retreatment (Figure 3). In the 1 participant who required 2 reinjections of ranibizumab (dark blue lines in Figure 3A), VSS peaked at both reinjection time points.

When using anti-VEGF treatment for ROP, close posttreatment monitoring is necessary to detect reactivation of the disease that requires retreatment.[12,25] It would be helpful to develop strategies for early identification of infants who have a high risk of ROP reactivation, particularly because late reactivations can occur months after treatment, and strict adherence to follow-up regimens can be difficult.[9-11,26,27] In this study, eyes with reactivated ROP had a larger change in VSS between baseline and 1 week after injection compared with eyes that did not have reactivated ROP in the subsequent study period (Figure 5). There are several potential explanations for this finding. One explanation is that there was a ceiling or floor effect, and eyes with higher VSS at baseline had more capacity to change after treatment. Previous work[19] found that higher VSS was associated with more posterior zone and higher stage and extent of stage 3 disease. We did not observe a statistically significant difference in VSS at baseline, but higher mean VSS was observed in eyes that later required retreatment, and the time of retreatment was earlier for eyes with higher VSS at baseline (Figure 3B). This finding is consistent with previous work[9] suggesting more posterior disease is more likely to reactivate. It is conceivable that this greater likelihood of reactivation may be explained by characteristics found in eyes that can change quickly; for instance, eyes with more extensive or flat neovascularization that is faster to respond to anti-VEGF treatment compared with eyes with more classic stage 3 disease may be more likely to reactivate. Our findings suggested that eyes that later reactivated were more likely to have higher VSS at baseline and lower VSS at week 1 than eyes that did not reactivate later (Figure 4); these results could support this hypothesis (ie, eyes that change quickly have characteristics that make them more likely to reactivate) but need to be further evaluated in future studies. In addition, the significant correlation between time of reactivation and baseline VSS observed in this study will have to be validated in a larger sample. If confirmed, such a potential estimate of when to expect a reactivation would have important implications in clinical practice and may help improve scheduling of follow-up visits after anti-VEGF treatment.

Strengths and Limitations

This study has strengths. All data were obtained from the CARE-ROP study, a well-documented prospective randomized clinical trial. In addition, more than 1000 images were available for analysis.

This study also has limitations. The main limitation is the small number of eyes available for analysis, and the results from this pilot study need to be further investigated in larger studies. In a larger study population, alternative analytic strategies, such as time to retreatment with VSS as a time-varying covariate or logistic modeling including treatment group as a covariate, could be performed. When interpreting VSS at the time of retreatment, it must be considered that the decision to provide retreatment during the study was not based on a VSS threshold but was determined clinically; thus, the VSS contains a subjective component from the treating physicians. Another limitation is that images were obtained only at prespecified intervals. Therefore, no images were available from time points before baseline injection (ie, the primary ROP screening phase), between week 4 and week 24 (except for participants requiring retreatment), or beyond week 24. The lack of photographic documentation during these periods limits assessment of disease trajectories but suggests that consistent photographic documentation and assessment of vascular severity might help with monitoring of disease reactivation. This documentation and assessment is important because ophthalmoscopy with scleral depression to visualize peripheral disease becomes increasingly challenging as children grow older. If future work confirms that monitoring for vascular severity near the posterior pole is highly sensitive to the presence of peripheral reactivation, then monitoring for reactivation using photography may supplement the challenging ophthalmoscopic examination. It has to be considered that the i-ROP algorithm was trained on images captured by a

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specific type of ophthalmic camera (RetCam), and the same type of camera was also used during the CARE-ROP trial. Generalizability to other fundus cameras will need to be evaluated.[28]

Conclusions

The results of this prognostic study suggested that the use of an AI-derived VSS may have utility in the detection of disease reactivation after anti-VEGF treatment for ROP. The data further suggested that the magnitude of early response to treatment (between baseline and week 1) may be associated with the risk of later ROP reactivation and, in eyes requiring retreatment, VSS at baseline might be used to estimate the time of retreatment. The explanation for these findings as well as their validation will be the target of future studies. If confirmed, an improved ability to identify infants at high risk of ROP reactivation and the ability to individually estimate the most likely time window for such a reactivation may improve the safety profile of ROP treatment.

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Author Contributions: Dr Stahl had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Drs Campbell and Stahl contributed equally as co-senior authors.

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Acquisition, analysis, or interpretation of data: Eilts, Pfeil, Poschkamp, Eter, Barth, Guthoff, Lagrèze, Grundel, Bründer, Busch, Chiang, Chan, Coyner, Ostmo, Campbell, Stahl.

Drafting of the manuscript: Eilts, Pfeil, Bründer, Campbell, Stahl.

Critical revision of the manuscript for important intellectual content: Pfeil, Poschkamp, Krohne, Eter, Barth, Guthoff, Lagrèze, Grundel, Busch, Kalpathy-Cramer, Chiang, Chan, Coyner, Ostmo, Campbell, Stahl.

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Supervision: Pfeil, Stahl.

Conflict of Interest Disclosures: Dr Pfeil reported receiving personal fees from Novartis Germany outside the submitted work. Dr Krohne reported receiving grants from Bayer and personal fees from Alimera Sciences, Allergan, Bayer, Novartis Germany, and Roche outside the submitted work. Dr Eter reported receiving grants from Bayer and personal fees from Alcon, Allergan, Apellis Pharmaceuticals, and Roche outside the submitted work. Dr Barth reported serving on the 2019 advisory board of Novartis Germany outside the submitted work. Dr Guthoff reported receiving grants from Bayer and personal fees from DORC outside the submitted work. Dr Busch

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reported receiving grants and personal fees from Bayer outside the submitted work. Dr Kalpathy-Cramer reported receiving grants from the National Institutes of Health during the conduct of the study and receiving grants from GE Healthcare and Genentech and serving as a consultant for Siloam Vision outside the submitted work. Dr Chiang reported receiving grants from Genentech and personal fees from Novartis Germany during the conduct of the study and owning equity in Inteleretina Professional Group outside the submitted work. Dr Chan reported receiving personal fees from Alcon and Genentech and being the owner of Siloam Vision outside the submitted work. Dr Coyner reported receiving personal fees from Boston AI Labs outside the submitted work. Dr Campbell reported receiving personal fees from Boston AI Labs and nonfinancial support from Siloam Vision during the conduct of the study and grants from Genentech outside the submitted work. Dr Stahl reported receiving personal fees from Novartis Germany and serving as a board member and speaker of the German Retina Society (Retinologische Gesellschaft) during the conduct of the study and receiving grants from Bayer and Novartis Germany and personal fees from Alcon, Apellis Pharmaceuticals, Bayer, Novartis Germany, and Roche outside the submitted work. No other disclosures were reported.

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Nonauthor Collaborators: The Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity (CARE-ROP) Study Group is listed in Supplement 2.

Data Sharing Statement: See Supplement 3.

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SUPPLEMENT 1.

eFigure 1. Settings for Fundus Images Taken During the CARE-ROP Trial
eFigure 2. Flowchart of Images and Patients Included in the Study

SUPPLEMENT 2.

Nonauthor Collaborators

SUPPLEMENT 3.

Data Sharing Statement

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